

A Self-Adjoint Operator Whose Norm Is Not Attained

Literature-implied status note for arXiv:2004.05496

9 August 2026

Status. Literature-implied answer, full for Open Question 1. The intended norm-attainment question has an affirmative answer already implicit in arXiv:1209.1218. This is not an original pipeline result.

Source question

N. B. Okelo, *Various notions of norm-attainability in normed spaces*, arXiv:2004.05496, Section 7, p. 12, asks [1]:

Let H be a reflexive, dense, separable, infinite dimensional complex Hilbert space. Does there exist a bounded self-adjoint operator $A : H \rightarrow H$ such that $\|Ax_0\| < \|A\|$ for all $x_0 \in H$?

Definition 2.5 of the same paper (p. 2) defines norm attainment using a *unit* vector. Thus the mathematically intended question is whether a bounded self-adjoint operator can satisfy

$$\|Ax\| < \|A\| \quad (\|x\| = 1).$$

Identification

S. Shkarin, *Norm attaining operators and pseudospectrum*, arXiv:1209.1218, p. 2, explicitly considers the diagonal operator D on ℓ_2 with diagonal entries

$$d_n = 1 - 2^{-n}, \quad n \in \mathbb{N},$$

and states that D has norm 1 which is not attained [2]. This is exactly the requested example. Indeed, $0 < d_n < 1$ and the diagonal is real, so D is bounded, positive, and self-adjoint. Moreover,

$$\|D\| = \sup_n d_n = 1,$$

whereas for every unit vector $x = (x_n)$,

$$\|Dx\|^2 = \sum_{n=1}^{\infty} d_n^2 |x_n|^2 < \sum_{n=1}^{\infty} |x_n|^2 = 1.$$

The strict inequality holds because at least one x_n is nonzero and every d_n^2 is strictly below 1. Every separable infinite-dimensional complex Hilbert space is unitarily isomorphic to ℓ_2 , so conjugating D by such a unitary gives the example on any H in the question.

Literal wording

Without the unit-vector restriction, the displayed quantifier in the source question is impossible. If $A \neq 0$, choose u with $Au \neq 0$ and then choose $t > \|A\|/\|Au\|$; this gives $\|A(tu)\| > \|A\|$. If $A = 0$, the requested strict inequality is $0 < 0$. Thus the literal reading has a negative answer, while the intended norm-attainment reading has the affirmative answer above.

Provenance, search evidence, and scope

Shkarin's 2012 paper predates Okelo's 2020 question and does not identify itself as answering that later question. The relation is therefore an agent-identified implication and belongs in the literature-implied category. An exact-phrase web search for Okelo's question found the source paper but no later paper explicitly advertising an answer; the run's local full-text index then located Shkarin's earlier diagonal example. The example is stated on p. 2 of the supporting PDF and the source question is on p. 12 of the source PDF.

This note fully answers Open Question 1 under both possible readings. It does not address Open Question 2 about when p -norm-attainability and p -normality coincide in general Banach spaces.

Human-review recommendation. Confirm that Okelo's Definition 2.5 supplies the unit-sphere convention and that Shkarin's diagonal operator is being considered over the complex field. No computational dependency is present.

References

- [1] N. B. Okelo, *Various notions of norm-attainability in normed spaces*, arXiv:2004.05496 (2020), Definition 2.5 and Section 7.
- [2] S. Shkarin, *Norm attaining operators and pseudospectrum*, arXiv:1209.1218 (2012), p. 2.